



Research Paper Presentation of IOHE

On

Quantitative Analysis and Algorithmic Trading in Financial Markets

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Introduction

Why Stock Market Prediction?

Stock markets are critical for investment and capital creation, yet highly unpredictable due to complex economic factors and constant fluctuations.

Limitations of Traditional Methods

Human-driven methods are tedious, rigid, and prone to emotional bias — particularly ill-suited during periods of high volatility.

Our Solution

An automated stock price prediction system using NIFTY 50 historical data with advanced ML/DL algorithms and data preprocessing pipelines.

50

NIFTY Stocks Analyzed

6+

ML/DL Models Compared

LSTM

Top Performing Model

0.897

Best R^2 Score (LSTM)

Problem Statement

Market Challenges

- Fail to perform consistently across different market conditions
- Generate many false signals
- Lead to false signals & reduced profitability
- Don't adapt to changing market
- Fixed strategies are ineffective in dynamic financial markets

Limitations of Traditional Strategies

- EMA crossover are simple and widely used
- Rule-based approaches treat all signals equally
- Unable to capture complex nonlinear patterns in financial data
- Ignore temporal dependencies in time-series data

Data & Model Gaps

- Need for a data-driven and adaptive approach
- Machine learning and deep learning can improve prediction accuracy
- Helps filter low-quality trades and enhance overall trading performance



Need for a data-driven and adaptive approach using Machine Learning and Deep Learning

Objectives

- ✓ Develop a data-driven stock prediction system for NIFTY 50
- ✓ Use ML & DL techniques for accurate forecasting
- ✓ Analyze historical data to identify patterns and trends
- ✓ Perform feature engineering with returns, lags & technical indicators
- ✓ Implement & compare multiple models
- ✓ Evaluate using RMSE, MAE & R^2 metrics
- ✓ Backtest strategies to analyze profitability
- ✓ Reduce false signals & improve trade selection
- ✓ Enhance prediction accuracy & model robustness
- ✓ Support intelligent, data-driven decision-making

Methodology



1

Data Acquisition

yFinance API
NIFTY50 OHLCV data

2

Preprocessing

Remove nulls
Min-Max scaling
Daily returns calc

3

Feature Engineering

Lag features
Statistical features
Return values

4

Train / Test Split

80% Training
20% Testing
Time-based split

5

Model Training

LR, XGBoost, LSTM
ARIMA, Prophet
Random Forest, SVR

6

Backtesting & Eval

Buy/Sell signals
MAE, MSE, RMSE
 R^2 & directional acc.

Feature Engineering



Price-Based Features

- Daily Return: $(\text{Close}_t - \text{Close}_{t-1}) / \text{Close}_{t-1}$
- Lag Features: 1-day, 3-day, 5-day price lags
- Rolling Mean: 7-day & 21-day moving averages
- Rolling Std Dev: 7-day & 21-day volatility

Volume Features

- Volume change % day-over-day
- Volume moving average (5-day)
- Volume vs price divergence indicator

Technical Indicators

- RSI (Relative Strength Index)
- MACD (Moving Avg Convergence Divergence)
- Bollinger Bands (Upper / Lower / Width)

Correlation Insight

Open, High, Low, Close are perfectly correlated ($r = 1.00$), showing multicollinearity.

Volume has weak negative correlation ($r = -0.27$) — useful as an independent signal.

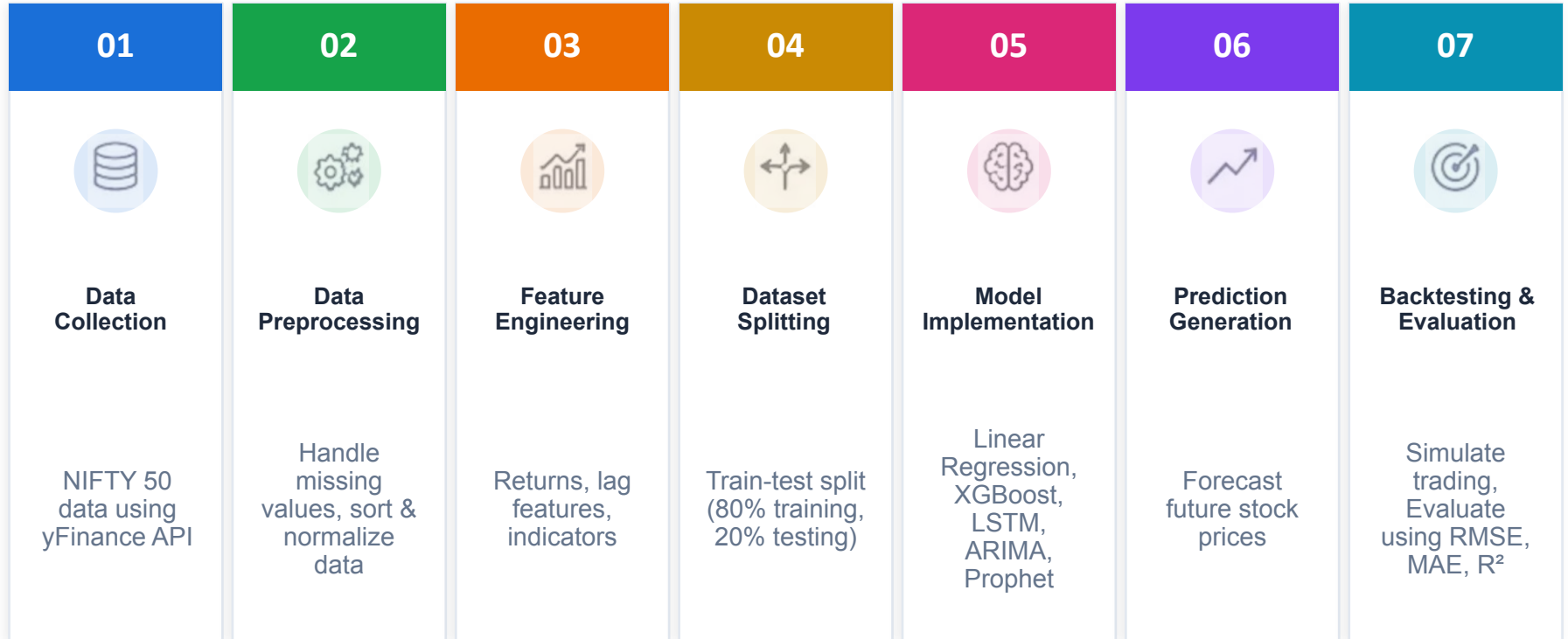
Normalization

Min-Max Scaling applied to standardize all features to $[0, 1]$ range before model training.

80% data used for training, 20% for testing (time-based split).

Total: 15+ engineered features per stock

Implementation Steps



Model Used



Linear Regression

Classical

Baseline model; low accuracy for volatile data.

Ridge Regression

Classical

Regularized variant; good for certain stocks.

Random Forest

Ensemble

Moderate performance; handles non-linearity.

XGBoost / Gradient Boosting

Ensemble

Strong for trend-based predictions; moderate accuracy.

SVR

Kernel

Moderate efficiency; handles complex boundaries.

LSTM

Deep Learning

Best overall; captures temporal dependencies. $R^2=0.897$, RMSE=41.69.

ARIMA

Statistical

Poor in volatile markets; negative R^2 values.

Prophet

Statistical

Ineffective for stock data; negative R^2 values.

Model Performance Metrics

Model	RMSE	MAE	R ²	Directional Acc.	Verdict
LSTM (TCS.NS)	89.81	89.12	0.8977	—	Best
SVR	59.26	65.12	0.8804	0.503	Good
Gradient Boosting	82.07	71.24	0.8805	0.499	Good
Random Forest	94.46	73.75	0.8868	0.444	Moderate
Ridge Regression	66.05	75.43	0.8810	0.439	Moderate
XGBoost	98.83	75.79	0.8811	0.499	Moderate
k-NN Regression	103.03	82.28	0.8641	0.433	Low
Linear Regression	104.26	81.99	0.8620	0.433	Low
ARIMA	312.46	329.72	-0.241	—	Poor
Prophet	317.61	295.99	-0.292	—	Poor

Results & Performance



LSTM achieved the highest accuracy ($R^2 = 0.96$, RMSE = 41.69)



Outperformed traditional models like ARIMA and Prophet



XGBoost performed well, especially on smaller datasets



Stock returns exhibited high volatility and fat-tailed distribution



Backtesting confirmed effective trading signals



Overall improvement in prediction accuracy

Model Comparison (R^2 Score)



LSTM Outperformed All Models

Results: Model Comparison (TCS.NS)

Model	RMSE	MAE	R ²	Dir. Acc.
LSTM (TCS.NS)	89.81	89.12	0.8977	—
SVR	59.26	65.12	0.8804	0.503
Gradient Boosting	82.07	71.24	0.8805	0.499
Random Forest	94.46	73.75	0.8868	0.444
Ridge	66.05	75.43	0.8810	0.439
XGBoost	98.83	75.79	0.8811	0.499
Ensemble (Ridge+RF+GB+ET+XGB)	98.83	82.19	0.8861	0.444
k-NN Regression	103.03	82.28	0.8641	0.433
Linear Regression	104.26	81.99	0.8620	0.433
ARIMA (TCS.NS)	312.46	329.72	-0.241	—
Prophet (TCS.NS)	317.61	295.99	-0.292	—

Conclusion

1 We developed a data-driven stock prediction system using ML & DL techniques for NIFTY 50.

2 LSTM captured time-series patterns effectively and outperformed traditional methods.

3 The study shows financial markets are complex and require advanced models.

4 Backtesting validated the effectiveness of our models in generating profitable trading signals.

5 The proposed approach improves prediction accuracy, reduces false signals, and supports intelligent, data-driven trading in dynamic markets.



$R^2 = 0.8977$

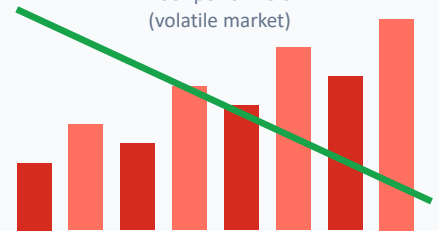
LSTM Best Model

RMSE = 41.69

TCS.NS Dataset

ARIMA $R^2 = -0.24$
Prophet $R^2 = -0.28$

Poor performers
(volatile market)



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THANK YOU